Brandon G. Oltarsh

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Education

Purdue University — Graduating December 2027
Major: Aerospace Engineering | **Minor:** Mathematics
GPA: 3.99

Experience

MFET 16300/10301 Teaching Assistant — Purdue University *September 2025 – Present*

- Collaborated with a team of 8 graders using Siemens Teamcenter and NX to evaluate student work.
- Provided feedback and grades to 1,200+ students, ensuring consistency and accuracy.

Cooling Analysis & Engine Optimization Member — Purdue Space Program (Active Controls) *2024 – 2025*

- Assisted in developing MATLAB-based regenerative cooling analysis for a 500-lbf liquid rocket engine.
- Designed methodology to evaluate film cooling efficiency and material performance.
- Researched and compared alcohol-based propellants and metals to optimize efficiency and thermal management.

Mechanical Director — FRC Robotics Team 2383 *Jan 2022 – Apr 2024*

- Led workshops and design sessions to train 40+ underclassmen in mechanical design, robot assembly, and workplace maintenance.
- Oversaw robotics lab and machine shop operations supporting 60+ members, boosting productivity.
- Designed and built competition-ready robots, contributing to 10 award-winning competition and outreach events.
- Collaborated with peers to prepare and present for team recognition and community awards.

Senior Counselor & Design Instructor — Robotics Summer Camp *Summers 2022 – 2024*

- Taught 50+ students (ages 10–14) mechanical design principles and CAD using Onshape.
- Created lasting hands-on challenges, lectures, and robot base designs to reinforce engineering concepts.
- Coordinated daily logistics for 100+ campers across multiple camp programs, ensuring safety and engagement.

Volunteering *Aug 2018 – Present*

- Led large community donation drives, securing 750+ of pounds of food for local homeless population.
- Packaged and delivered food and clothing to 200+ food-insecure families in Broward County Florida.
- Created a gun safety initiative, presenting at multiple community events and distributing 800+ specialty gun locks.

Projects

BMW 2002 (1973) and BMW 1600 (1969) Restoration

- Built a rotisserie to perform body work, including laser rust removal and MIG welding.
- Repaired taillight wiring and shift knob functionality for BMW 1600.
- Next steps: full mechanical reassembly, interior rebuild, and paint.

Derbi Variant (moped) Repairs

- Designed a #41 chain sprocket to be cut by water jet for moped chain drive.
- Rebuilt air intake for CVT 2-Stroke engine.

Casting/Smithing forge

- Built a casting furnace and top-down smithing forge for working with aluminum, brass, silver, iron, and steel.

Skills

Engineering Tools: NX, Siemens PDM/PLM, Teamcenter, Onshape, Autodesk Inventor, CAM, CNC Machining, FDM 3D Printing, IBM SPSS Statistics, Solidworks
Programming: MATLAB, Python, HTML, CSS, C
Productivity: Excel, Word, PowerPoint, Google and Microsoft Suite